Z3 Option 1

```
s SATISFIABLE
v 1 -2 -3 -4 -5 -6 -7 -8 -9 -10 -11 -12 -13 -14 -15 16 -17 -18 -19 -20 -21 -22 -23 -24 -25 -26 -27 -28 -29 -30 -31 -32 -33
-34 -35 -36 -37 -38 -39 -40 -41 42 -43 -44 -45 -46 -47 -48 -49 -50 -51 -52 -53 -54 -55 -56 -57 -58 -59 -60 -61 -62 -63 -64
-65 -66 -67 -68 -69 -70 71 -72 -73 -74 -75 -76 -77 -78 -79 -80 81 -82 -83 -84 -85 -86 -87 -88 -89 -90 -91 -92 -93 -94 -95 9
6 -97 -98 -99 -100 -101 -102 -103 -104 -105 -106 -107 -108 -109 110 -111 -112 -113 -114 -115 -116 -117 -118 -119 -120 -121
-122 -123 -124 -125 -126 -127 -128 -129 -130 -131 -132 -133 -134 -135 -136 -137 -138 -139 -140 141 -142 -143 -144 -145 -146
 -147 -148 -149 -150 -151 -152 -153 -154 -155 -156 -157 -158 -159 -160 -161 -162 -163 164 -165 -166 -167 -168 -169 -170 -17
1 -172 -173 -174 -175 -176 -177 178 -179 -180 -181 -182 -183 -184 -185 -186 -187 -188 -189 -190 -191 -192 -193 -194 -195 0
```

Z3 option 2

```
s SATISFIABLE
v -1 2 -3 -4 -5 6 -7 -8 9 -10 11 -12 -13 -14 15 16 -17 -18 19 -20 -21 22 -23 -24 -25 -26 27 28 -29 -30 31 -32 -33 -34 -35 3
6 -37 -38 -39 -40 -41 42 -43 -44 -45 -46 -47 -48 -49 -50 -51 -52 -53 54 -55 -56 -57 -58 -59 60 61 -62 -63 -64 -65 -66 -67 -
68 -69 -70 -71 -72 -73 -74 -75 76 77 -78 -79 -80 -81 -82 -83 84 -85 -86 -87 -88 -89 90 -91 -92 -93 -94 -95 0
```

Minsat Option 1

```
============================[ Problem Statistics ]=============================
|                                                                             |
|  Number of variables:           195                                         |
|  Number of clauses:            3127                                         |
|  Parse time:                   0.01 s                                       |
|  Eliminated clauses:           0.00 Mb                                      |
|  Simplification time:          0.04 s                                       |
|                                                                             |
============================[ Search Statistics ]==============================
| Conflicts |          ORIGINAL         |          LEARNT          | Progress |
|           |    Vars  Clauses Literals |    Limit  Clauses Lit/Cl |          |
===============================================================================
===============================================================================
restarts              : 1
conflicts             : 0              (0 /sec)
decisions             : 115            (0.00 % random) (2333 /sec)
propagations          : 157            (3185 /sec)
conflict literals     : 0              (-nan % deleted)
Memory used           : 11.06 MB
CPU time              : 0.049293 s

SATISFIABLE
```

Minsat option 2

```
============================[ Problem Statistics ]=============================
|                                                                             |
|  Number of variables:           95                                          |
|  Number of clauses:            1396                                         |
|  Parse time:                   0.01 s                                       |
|  Eliminated clauses:           0.00 Mb                                      |
|  Simplification time:          0.03 s                                       |
|                                                                             |
============================[ Search Statistics ]==============================
| Conflicts |          ORIGINAL         |          LEARNT          | Progress |
|           |    Vars  Clauses Literals |    Limit  Clauses Lit/Cl |          |
===============================================================================
===============================================================================
restarts              : 1
conflicts             : 0              (0 /sec)
decisions             : 44             (0.00 % random) (1343 /sec)
propagations          : 47             (1435 /sec)
conflict literals     : 0              (-nan % deleted)
Memory used           : 11.06 MB
CPU time              : 0.032758 s

SATISFIABLE
```

Kissat option 1

```
c Kissat SAT Solver
c
c Copyright (c) 2021-2024 Armin Biere University of Freiburg
c Copyright (c) 2019-2021 Armin Biere Johannes Kepler University Linz
c
c Version 4.0.4 8af8e56f174b778aef3aa45af9f739b2a5f492c2
c gcc (Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0 -W -Wall -O3 -DNDEBUG
c Tue Mar 10 02:09:46 IST 2026 Linux the-nightsky 6.8.0-94-generic x86_64
c
c ---- [ parsing ] ------------------------------------------------------------
c
c opened and reading DIMACS file:
c
c   option1.cnf
c
c parsed 'p cnf 195 3127' header
c closing input after reading 35086 bytes (34 KB)
c finished parsing after 0.00 seconds
c
c ---- [ solving ] ------------------------------------------------------------
c
c seconds switched rate     size/glue tier1  binary    remaining
c        MB reductions conflicts size  tier2  irredundant
c         level restarts redundant glue  trail     variables
c
c *  0.00  3 0 0 0  0 0   0   0 0.0 0 0 0 0 0% 3117 10 195 100%
c lucky in forward setting literals to false
c l  0.00  4 0 0 0  0 0   0   0 0.0 0 0 0 0 0% 3117 10 195 100%
c 1  0.00  4 0 0 0  0 0   0   0 0.0 0 0 0 0 0% 3117 10 195 100%
c
c ---- [ result ] ------------------------------------------------------------
c
s SATISFIABLE
v -1 -2 -3 -4 -5 -6 -7 -8 -9 -10 -11 -12 -13 -14 15 -16 -17 -18 -19 -20 -21
v -22 -23 -24 -25 -26 -27 -28 -29 -30 -31 -32 33 -34 -35 -36 -37 -38 -39 -40
v -41 -42 -43 -44 -45 -46 -47 -48 -49 -50 -51 -52 -53 -54 -55 -56 -57 -58 -59
v 60 -61 -62 -63 -64 -65 -66 -67 -68 -69 70 -71 -72 -73 -74 -75 -76 -77 -78
v -79 -80 -81 -82 -83 -84 -85 -86 -87 -88 -89 90 -91 -92 -93 -94 -95 -96 -97
v -98 -99 -100 -101 -102 103 -104 -105 -106 -107 -108 -109 -110 -111 -112 -113
v -114 -115 -116 -117 -118 -119 -120 -121 -122 -123 -124 -125 -126 -127 -128
v -129 -130 -131 -132 -133 -134 -135 -136 -137 138 -139 -140 -141 -142 -143
v -144 -145 -146 -147 -148 -149 -150 -151 152 -153 -154 -155 -156 -157 -158
v -159 -160 -161 -162 -163 -164 165 -166 -167 -168 -169 -170 -171 -172 -173
v -174 -175 -176 -177 -178 -179 -180 -181 -182 -183 -184 -185 -186 -187 -188
v -189 -190 -191 192 -193 -194 -195 0
c
c ---- [ profiling ] ----------------------------------------------------------
c
c           0.00   77.46 %  parse
c           0.00    2.46 %  lucky
c           0.00    0.00 %  search
c           0.00    0.00 %  simplify
c ==============================================
c           0.00  100.00 %  total
```

Kissat option 2

```
c gcc (Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0 -W -Wall -O3 -DNDEBUG
c Tue Mar 10 02:09:46 IST 2026 Linux the-nightsky 6.8.0-94-generic x86_64
c
c ---- [ parsing ] ------------------------------------------------------------
c
c opened and reading DIMACS file:
c
c   option2.cnf
c
c parsed 'p cnf 95 1396' header
c closing input after reading 22859 bytes (22 KB)
c finished parsing after 0.00 seconds
c
c ---- [ solving ] ------------------------------------------------------------
c
c seconds switched rate     size/glue tier1 binary     remaining
c        MB reductions conflicts size  tier2  irredundant
c         level restarts redundant glue  trail     variables
c
c *  0.00  4 0 0 0  0 0   0   0 0.0 0 0 0 0 0% 227 1169 95 100%
c lucky in backward setting literals to true
c l  0.00  4 0 0 0  0 0   0   0 0.0 0 0 0 0 0% 227 1169 95 100%
c 1  0.00  4 0 0 0  0 0   0   0 0.0 0 0 0 0 0% 227 1169 95 100%
c
c ---- [ result ] ------------------------------------------------------------
c
s SATISFIABLE
v -1 2 -3 -4 -5 6 -7 -8 9 -10 -11 12 13 -14 -15 -16 17 -18 -19 -20 21 -22 -23
v 24 -25 26 -27 -28 -29 30 -31 -32 -33 -34 35 -36 -37 -38 -39 -40 41 -42 -43
v -44 -45 -46 -47 -48 -49 50 -51 -52 -53 -54 55 -56 -57 -58 -59 60 -61 -62 -63
v -64 65 -66 -67 -68 -69 -70 -71 -72 -73 -74 -75 76 -77 -78 -79 -80 -81 -82 83
v -84 -85 -86 -87 -88 89 -90 -91 -92 -93 -94 95 0
c
c ---- [ profiling ] ----------------------------------------------------------
c
c           0.00   74.06 %  parse
c           0.00    7.91 %  lucky
c           0.00    0.00 %  search
c           0.00    0.00 %  simplify
c ================================================
c           0.00  100.00 %  total
c
c ---- [ statistics ] --------------------------------------------------------
c
c conflicts:                                0                0.00 per second
c decisions:                                0                0.00 per conflict
c propagations:                           294           156134    per second
c switched:                                 0                0    interval
c
c ---- [ glue usage ] ---------------------------------------------------------
c
c no clauses used at all
c
c ---- [ resources ] ----------------------------------------------------------
c
c maximum-resident-set-size:          3801088 bytes          4 MB
c process-time:                                              0.00 seconds
```